

Anton Matiash

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Professional Summary

Mechanical Engineering undergrad (Top 1% cohort) with a strong background in experimental physics (IYPT Gold Medalist). Focused on sim-to-real transfer and reinforcement learning for bio inspired legged robotics. Passionate about bridging hardware and software through dynamic simulation, control theory, and modern machine learning techniques. Seeking a PhD position to tackle complex challenges in autonomous robotics.

Education

BEng in Mechanical Engineering

Sep 2024 – May 2027

Swansea University

Swansea, UK

First Year Average: 89%(Top 1 in cohort), Second Year Average: 86%(Top 1 in cohort)

Most important first year modules: Data Analysis and Simulation(98%), Engineering Design 1(95%), Engineering Mathematics(91%)

Second Year: Machines and Control (99%), AI, Statistics and Machine Learning(80%), Engineering Design 2(81%), Dynamics I(94%)

Modern Robotics specialisation (MR)

June 2025 – Oct 2025

Northwestern University/Coursera

Online

Completed all 6 courses in specialisation: Foundations of Robot Motion, Robot Kinematics, Dynamics, Motion Planning and Control, Manipulation and Wheeled Mobile Robots, Capstone Project.

High School Diploma

Sep 2017 – June 2024

Kharkiv Physics-Mathematics Science Lyceum №27

Kharkiv, Ukraine

Last year Average: 11.1 out of 12. Top 2 in Cohort

Experience

Research Intern

Jun 2026 – Aug 2026

Bristol Robotics Laboratory

Bristol, UK

- Quantified the feasibility of using Parallel Elastic Joints (PEJs) in robotic applications by conducting a series of simulations and experiments, demonstrating their potential for high energy efficiency.
- Iterated the training pipeline through multiple setbacks to minimize Simulation-to-Real gaps as much as possible.
- Compressed findings into a scientific paper, which will be submitted soon.

Makerspace Supervisor

Sep 2025 – Aug 2027

CREU: The Manufactory

Swansea, UK

- Helped manage a student makerspace, maintained 3D printers, laser cutters, and other equipment. Helped establish basic working procedures for the makerspace and trained new students on how to use the equipment safely.

Secretary

Jul 2026 – Jul 2027

Swansea University Table Tennis Club

Swansea, UK

- Handled the administrative tasks of the club, making sure that no one forgets important dates or deadlines.

Honors & Awards

International Physicists' Tournament 2025

Kielce, Poland

Achieved 4th place among 17 countries worldwide as part of Team Ukraine, was solving mechanics problem about staircase motion of "Slinky" with experiments, simulations and theory over the year period and presented solution at the tournament achieving highest score for the report among my team.

International Young Physicists' Tournament 2024

Budapest, Hungary

Achieved Gold Medal, competed among 38 countries worldwide, solved complex mechanics/fluids dynamics problem about soap film motion on complex shapes over the year period and presented it to the audience at the tournament achieving highest score for the report among my team.

All-Ukrainian Student Physics Olympiad 2023/2024

Ivano-Frankivsk, Ukraine

Achieved 3rd place Diploma on the national stage, solved 4 theoretical and 2 experimental problems over the period of 6 hours, 2 points were missing to qualify for IPhO (which is the international stage of the Olympiad).

Selected Projects

Quadruped Robot design and Reinforcement Learning (RL) Project *Oct 2025 – Present*

ROS, Robotics, Engineering, Fusion360, C++, IsaacSim, Python

- Designing and building a custom 12-DOF quadruped robot utilising BLDC motors and LIDAR for advanced control and navigation via ROS. Currently designed and tested rotational actuator with 7:1 gear ratio.
- Researching and Developing a comprehensive Sim-to-Real pipeline by constructing accurate physical models in Isaac Sim to facilitate safe policy training.
- Researching Reinforcement Learning algorithms in Python to train robust locomotion policies for uneven terrain, serving as the core framework for my upcoming independent thesis research.

Capstone Project (part of the MR specialisation)

Jun 2025 – Oct 2025

MATLAB, Robotics, Computer Science

[\[Certificate Link\]](#)[\[Demonstration Video\]](#)

- Engineered a complete kinematics and control software stack for a Kuka mobile manipulator in MATLAB, successfully executing automated pick-and-place tasks in CoppeliaSim.
- Developed a custom trajectory generator and PID controller using MATLAB to accurately manage joint limits and achieve precise end-effector positioning from randomized starting configurations.

IMechE Foundation Design Challenge

Jan 2025 – June 2025

Fusion360, Engineering, Physics(Electronics, Mechanics), Powerpoint

[\[Project poster\]](#)

- Designed and manufactured a car that could move forward to a point, stop, wait 15 seconds, and return to the start point, with purely analog control.
- Involved designing mechanical components, and custom electrical circuit (Complete information in poster).

Other Projects

Mafia Bot *TypeScript, Node.js, Discord.js, TypeORM* [\[GitHub\]](#), **Strategy Game** *C#, Godot Engine*, [\[GitHub\]](#)

Technical Skills

Engineering: Fusion 360 (CAD & CAM), Wolfram Mathematica, Granta EduPack, MATLAB Simulink, Isaac Sim.

Programming Languages: MATLAB, Wolfram Lang, C++, C#, TypeScript, Python, JS/HTML/CSS, SQL

Frameworks & Libraries: ROS, Pytorch, Isaac Lab

DevOps: Git, Docker, GitHub

Languages: English (C1), Russian (Native), Ukrainian (Native)

Other: LaTeX, Markdown, Microsoft Office